Fiscal Year:	FY 2013	**Task Last Updated:**	FY 04/28/2013
PI Name:	Rithidech, Kanokporn Ph.D.		
Project Title:	In vivo induction of chromosome instability and aberrant patterns of DNA methylation in hematopoietic stem/progenitor cells by heavy ions		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	11794-8691	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiobiology NNJ10ZSA001N
Start Date:	07/01/2011	**End Date:**	06/30/2015
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date is now 6/30/2015, per NSSC information (Ed., 4/22/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Whorton, Elbert Ph.D. (Galveston National Laboratory)		
Grant/Contract No.:	NNX11AK91G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Space environment consists of various types of radiation that are different from those found in the Earth's atmosphere. These include particles with high mass and high energy (heavy particles) such as silicon and titanium. It has been estimated that every cell in the body may be hit by these particles several times within a month during a journey to Mars. Hence, exposure to radiation in space is the greatest hazard to astronauts venturing beyond Earth. It is therefore important to protect astronauts in space environments. This will require improvement in our knowledge of radiation-induced changes in specific target cells that may impact the health of astronauts. Cancer of blood cells, in particular myeloid leukemia, is one of the major health concerns from exposure to space radiation. However, the predictions of risks for developing cancer of blood cells after exposure to space radiation are unsatisfactory. To increase the reliability of predicting risk for developing cancer of blood cells, a much improved understanding of space radiation-induced changes in the target cells (stem cells and their progenitors) in living organisms for the induction of this particular type of cancer is critically important. In this project, we exposed mice to two types of heavy ions found in space and relevant to NASA missions, i.e. silicon and titanium. We are conducting experiments to determine the effects of these ions on the induction of damage at the genetic level by measuring the frequency and type of initial- and delayed-occurring chromosome instability (normally known as genomic instability). Such analyses are being done in hematopoietic stem/progenitor cells (HSPCs) residing in the bone marrow (BM) of exposed mice collected at different times post-irradiation using a novel genome-wide multicolor fluorescence in situ hybridization (whole genome mFISH). The measurement of chromosome instability in HSPCs is coupled with the characterization of damage at the epigenetic level (i.e. the damage affecting gene function without a change in the sequence of DNA molecules) by determining the pattern of DNA-methylation. These two biological endpoints are highly relevant for assessing cancer risks; however, they have not been applied with in vivo exposure to space radiation with silicon or titanium ions. Hence, our results will provide high-priority information for NASA. Our research is the first to use integrative in vivo endpoints in the HSPCs that are the target of radiation-induced myeloid leukemia to evaluate the effects of space-radiation. Our data will not only contribute to reducing uncertainties in the assessment of cancer risk from exposure to radiation during space flights but will also improve our understanding of the molecular basis for the influence of space radiation on genomic instability and aberrant DNA-methylation profiles.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	Our data will provide important information that will lead to a better understanding of the patterns of radiation-induced genetic and non-genetic (epigenetic) damage in HSPCs that trigger the induction of genomic instability, a key event of carcinogenesis. This study will couple molecular mechanisms with quantitative data from a robust and highly sensitive technique. The resulting data will deliver information that is of high priority for NASA because the data will not only contribute to reducing the uncertainties in the assessment of cancer risk from exposure to radiation (heavy ions) during space flights but will also improve the understanding of the molecular basis for radiation-induced genomic instability. Hence, the obtained data are critically important for understanding risks associated with space travel as well as developing advanced radiation (heavy ions) therapies for cancer treatments on Earth. Moreover, the information obtained from this project will have a significant impact on the future development of a novel mechanism-based therapeutic strategy to prevent or mitigate the acute or long-term effects of radiation on cells at risk for myeloid leukemia induced by radiation found in space, as well as those found on Earth.		

Task Progress:	This project was initiated in July 2011. This is the second taskbook of our accomplishments of the past 1.5 years. The major goals during this period were to: 1. complete the exposure of mice to different doses of 28Si or 48Ti ions, 2. collect BM cells from sham controls and exposed mice, 3. obtain colonies of BM-derived HSPCs at different times up to six months after irradiation, 4. determine the number and types (according to morphology) of the BM-derived HSPC colonies, 5. collect BM-derived HSPCs obtained from sham controls and exposed mice for the analyses of chromosome aberrations (CAs) and DNA methylation induced in vivo, 6. measure the frequency of CAs and determine the pattern of DNA methylation in HSPCs obtained from sham controls and exposed mice. We have accomplished goals 1 to 5. With respect to goal #6, we have made substantial progress in cytogenetic and DNA methylation (5-methyl cytosine and its oxidized form, i.e. 5-hydroxymethyl cytosine or 5-hmC). For the cytogenetic analysis (by means of the mFISH method), our current focus is on the delayed- or late-occurring CAs because it is well recognized that it is the best-established end point of radiation-induced genomic instability (a fundamental mechanism known to elevate cancer risk). We have completed the analysis of CAs in HSPCs collected at 6 months after exposure of CBA/Ca mice to silicon ions. Several types of aberrations were observed in HSPCs collected from exposed CBA/CaJ mice. These include abnormal cells, chromatid breaks, chromosome breaks, and translocations (exchanges). We found dose-dependent increases in the frequencies of abnormal cells, chromatid and chromosome breaks (Analysis of Variance, p=0.0008, or 0.00005 or 0.02, respectively). Of note, our data also showed significant increases in the frequencies of exchanges in HSPCs collected from mice exposed to a medium (0.25 Gy) or a high (0.5 Gy) dose of silicon ions. Overall, delayed- or late-occurring chromosomal damage was clearly observed in HSPCs collected from mice exposed to silicon ions, suggesting the occurrence of genomic instability. Since genomic instability is a fundamental mechanism known to elevate cancer risk, our data provide high-priority and highly relevant information for NASA. Although the mFISH method is robust and capable of measuring the frequency, the degree of complexity, and the dose response for the production and persistence of simple and complex aberrations over time, it is very time consuming. Due to a very large number of metaphase cells to be scored, the evaluation of genome-wide CAs by mFISH is our current major effort and will be continued into year 3 of the study. Relating to the determination of the pattern of DNA methylation in HSPCs, we have made significant progress in determining the global (total) levels of 5-methyl cytosine and 5-hydroxymethyl cytosine in DNA isolated from HSPCs using an ELISA-based method. Such analyses were completed in HSPCs obtained from cells collected at 1 wk, 1 mo, and 6 mos after exposure of mice to silicon or titanium ions. Our data suggest that the global level of 5-hydroxymethyl cytosine may be a better predictor of exposure to HZE particles found in space that, in turn, might be more useful for risk assessment than the global level of 5-methyl cytosine. We are currently measuring the expression of enzymes known to be involved in the formation of DNA methylation. Ultimately, we will assess if the global levels of DNA methylation are correlated with chromosomal instability, as a function of silicon- titanium-ion dose and time post-irradiation.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
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